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Chapter 6

Sets

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Exercise 6(A)

Question 1(i)

$\{x : x \in \mathbb{Z} \text{ and } x^2 - 4x = 0\}$ is equal to :

1. $\{4\}$
2. $\{0, 4\}$
3. $\{0, 2\}$





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Answer

$$x^2 - 4x = 0$$

$$\Rightarrow x(x - 4) = 0$$

$$\Rightarrow x = 0 \text{ or } (x - 4) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 4$$

$$\{x : x \in \mathbb{Z} \text{ and } x^2 - 4x = 0\} = \{0, 4\}$$

Hence, option 2 is the correct option.**Question 1(ii)** $\{x : x \in \mathbb{Z} \text{ and } x^2 = 9\}$ is equal to :

1. $\{3\}$

2. $\{0, 3\}$

3. $\{0, -3\}$

4. $\{-3, 3\}$

Answer

$$x^2 = 9$$

$$\Rightarrow x = \sqrt{9}$$

$$\Rightarrow x = 3 \text{ or } -3$$

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$$\{x : x \in \mathbb{Z} \text{ and } x^2 = 9\} = \{-3, 3\}$$

Hence, option 4 is the correct option.

Question 1(iii)

If set $A = \{x : x \in \mathbb{N}, x = n - 3, n \in \mathbb{N} \text{ and } n < 3\}$, then set A is :

1. $\{-3, -2, -1, 0, 2\}$

2. $\{\}$

3. $\{-3, -2, -1\}$

4. $\{0, -1, -2, -3\}$

Answer

$$x = n - 3$$

When $n = 0$

$$x = 0 - 3 = -3$$

When $n = 1$

$$x = 1 - 3 = -2$$

When $n = 2$

$$x = 2 - 3 = -1$$

As $-3, -2, -1$ are not natural numbers,

$$\therefore \text{set } A = \{x : x \in \mathbb{N}, x = n - 3, n \in \mathbb{N} \text{ and } n < 3\} = \{\}$$

Hence, option 2 is the correct option.

Question 1(iv)

Set $\{x : x = n^2 - 1, x \in \mathbb{Z} \text{ (integers) and } 2 < n \leq 5\}$ is equal to :

1. $\{8, 15, 24\}$
2. $\{3, 8, 15\}$
3. $\{3, 8, 15, 24\}$
4. none of these

Answer

$$x = n^2 - 1$$

When $n = 3$

$$\Rightarrow x = 3^2 - 1$$

$$\Rightarrow x = 9 - 1 = 8$$

When $n = 4$

$$\Rightarrow x = 4^2 - 1$$

$$\Rightarrow x = 16 - 1 = 15$$

When $n = 5$

$$x = 5^2 - 1$$

$$\Rightarrow x = 25 - 1 = 24$$

$$\{x : x = n^2 - 1, x \in \mathbb{Z} \text{ (integers) and } 2 \leq n \leq 5\} = \{8, 15, 24\}$$

Hence, option 1 is the correct option.

Question 1(v)

Set-builder form of set $A = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right\}$ is :

1. $\left\{x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } n > 1\right\}$
2. $\left\{x = \frac{n-1}{n-2}, n \in \mathbb{N} \text{ and } n > 2\right\}$
3. $\left\{x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 < n < 2\right\}$
4. $\left\{x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 \leq n < 4\right\}$

Answer

$$\begin{aligned} \text{set } A &= \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right\} \\ &= \left\{\frac{1}{1+1}, \frac{2}{2+1}, \frac{3}{3+1}\right\} \\ &= \left\{x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 \leq n < 4\right\} \end{aligned}$$

Hence, option 4 is the correct option.

Question 2(i)

Write the following set in roster (Tabular) form :

$$A_1 = \{x : 2x + 3 = 11\}$$

Answer

$$A_1 = \{x : 2x + 3 = 11\}$$

$$\Rightarrow 2x = 11 - 3$$

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = \frac{8}{2}$$

$$\Rightarrow x = 4$$

$$A_1 = \{4\}$$

Question 2(ii)

Write the following set in roster (Tabular) form :

$$A_2 = \{x : x^2 - 4x - 5 = 0\}$$

Answer

$$A_2 = \{x : x^2 - 4x - 5 = 0\}$$

$$\Rightarrow x^2 - 5x + x - 5 = 0$$

$$\Rightarrow x(x - 5) + 1(x - 5) = 0$$

$$\Rightarrow (x - 5)(x + 1) = 0$$

$$\Rightarrow (x - 5) = 0 \text{ or } (x + 1) = 0$$

$$\Rightarrow x = 5 \text{ or } x = -1$$

$$A_2 = \{5, -1\}$$

Question 2(iii)

Write the following set in roster (Tabular) form :

$$A_3 = \{x : x \in \mathbb{Z}, -3 \leq x < 4\}$$

Answer

$$A_3 = \{x : x \in \mathbb{Z}, -3 \leq x < 4\}$$

$$= \{x : x \in \mathbb{Z}, -3 \leq x < 4\}$$

Numbers between -3 and 4 are -3, -2, -1, 0, 1, 2, 3

$$A_3 = \{-3, -2, -1, 0, 1, 2, 3\}$$

Question 2(iv)

Write the following set in roster (Tabular) form :

$$A_4 = \{x : x \text{ is a two digit number and sum of the digits of } x \text{ is } 7\}$$

Answer

$$A_4 = \{x : x \text{ is a two digit number and sum of the digits of } x \text{ is } 7\}$$

$$16 \Rightarrow 1 + 6 = 7$$

$$25 \Rightarrow 2 + 5 = 7$$

$$34 \Rightarrow 3 + 4 = 7$$

$$43 \Rightarrow 4 + 3 = 7$$

$$52 \Rightarrow 5 + 2 = 7$$

$$61 \Rightarrow 6 + 1 = 7$$

$$70 \Rightarrow 7 + 0 = 7$$

$$\mathbf{A_4 = \{16, 25, 34, 43, 52, 61, 70\}}$$

Question 2(v)

Write the following set in roster (Tabular) form :

$$A_5 = \{x : x = 4n, n \in W \text{ and } n < 4\}$$

Answer

$$A_5 = \{x : x = 4n, n \in W \text{ and } n < 4\}$$

When $n = 0$

$$\Rightarrow x = 4 \times 0 = 0$$

When $n = 1$

$$\Rightarrow x = 4 \times 1 = 4$$

When $n = 2$

$$\Rightarrow x = 4 \times 2 = 8$$

When $n = 3$

$$\Rightarrow x = 4 \times 3 = 12$$

$$\mathbf{A_5 = \{0, 4, 8, 12\}}$$

Question 2(vi)

Write the following set in roster (Tabular) form :

$$A_6 = \left\{ x : x = \frac{n}{n+2}; n \in N \text{ and } n > 5 \right\}$$

Answer

$$A_6 = \left\{ x : x = \frac{n}{n+2}; n \in N \text{ and } n > 5 \right\}$$

When $n = 6$

$$\Rightarrow \frac{6}{6+2} = \frac{6}{8}$$

When $n = 7$

$$\Rightarrow \frac{7}{7+2} = \frac{7}{9}$$

When $n = 8$

$$\Rightarrow \frac{8}{8+2} = \frac{8}{10}$$

When $n = 9$

$$\Rightarrow \frac{9}{9+2} = \frac{9}{11}$$

$$A_6 = \left\{ x : x = \frac{6}{8}, \frac{7}{9}, \frac{8}{10}, \frac{9}{11}, \dots \right\}$$

$$A_6 = \left\{ \frac{3}{4}, \frac{7}{9}, \frac{4}{5}, \frac{9}{11}, \dots \right\}$$

Question 3(i)

Write the following set in set-builder (Rule Method) form :

$$B_1 = \{6, 9, 12, 15, \dots\}$$

Answer

$$B_1 = \{6, 9, 12, 15, \dots\}$$

$$B_1 = \{3 \times 1 + 3, 3 \times 2 + 3, 3 \times 3 + 3, 3 \times 4 + 3, \dots\}$$

$$B_1 = \{x : x = 3n + 3, n \in \mathbb{N}\}$$

Question 3(ii)

Write the following set in set-builder (Rule Method) form :

$$B_2 = \{11, 13, 17, 19\}$$

Answer

$$B_2 = \{11, 13, 17, 19\}$$

$$B_2 = \{x : x \text{ is a prime number } 10 < x < 20\}$$

Question 3(iii)

Write the following set in set-builder (Rule Method) form :

$$B_3 = \left\{ \frac{1}{3}, \frac{3}{5}, \frac{5}{7}, \frac{7}{9}, \frac{9}{11}, \dots \right\}$$

Answer

$$B_3 = \left\{ \frac{1}{3}, \frac{3}{5}, \frac{5}{7}, \frac{7}{9}, \frac{9}{11}, \dots \right\}$$

$$B_3 = \left\{ \frac{1}{1+2}, \frac{3}{3+2}, \frac{5}{5+2}, \frac{7}{7+2}, \frac{9}{9+2}, \dots \right\}$$

$$B_3 = \left\{ \frac{n}{n+2}; \text{where } n \text{ is an odd natural number} \right\}$$

Question 3(iv)

Write the following set in set-builder (Rule Method) form :

$$B_4 = \{8, 27, 64, 125, 216\}$$

Answer

$$B_4 = \{8, 27, 64, 125, 216\}$$

$$B_4 = \{2^3, 3^3, 4^3, 5^3, 6^3\}$$

$$B_4 = \{x : x = n^3; n \in \mathbf{N} \text{ and } 2 \leq n \leq 6\}$$

Question 3(v)

Write the following set in set-builder (Rule Method) form :

$$B_5 = \{-5, -4, -3, -2, -1\}$$

Answer

$$B_5 = \{-5, -4, -3, -2, -1\}$$

$$B_5 = \{n; n \in \mathbf{Z} \text{ and } -5 \leq n \leq -1\}$$

Question 3(vi)

Write the following set in set-builder (Rule Method) form :

$$B_6 = \{ \dots, -6, -3, 0, 3, 6, \dots \}$$

Answer

$$B_6 = \{ \dots, -6, -3, 0, 3, 6, \dots \}$$

$$B_6 = \{ \dots, 3 \times (-2), 3 \times (-1), 3 \times 0, 3 \times 1, 3 \times 2, \dots \}$$

$$B_5 = \{x : x = 3n; n \in \mathbb{Z}\}$$

Question 4(i)

Is $\{1, 2, 4, 16, 64\} = \{x : x \text{ is a factor of } 32\}$? Give reason.

Answer

No, as 64 is not a factor of 32. So this two given sets are unequal.

Hence, $\{1, 2, 4, 16, 64\} \neq \{x : x \text{ is a factor of } 32\}$

Question 4(ii)

Is $\{x : x \text{ is a factor of } 27\} \neq \{3, 9, 27, 54\}$? Give reason.

Answer

Yes, as 54 is not a factor of 27. So the two given sets are unequal.

Hence, $\{x : x \text{ is a factor of } 27\} \neq \{3, 9, 27, 54\}$

Question 4(iii)

Write the set of even factors of 124.

Answer

Factors of 124 = 1, 2, 4, 31, 62, 124

Even factors of 124 = 2, 4, 62, 124

{2, 4, 62, 124}

Question 4(iv)

Write the set of odd factors of 72.

Answer

Factors of 72 = 1, 2, 3, 4, 6, 8, 9, 12, 18, 36, 72

Odd factors of 72 = 1, 3, 9

{1, 3, 9}

Question 4(v)

Write the set of prime factors of 3234.

Answer

Factors of 3234 = 1, 2, 3, 6, 7, 11, 14, 21, 22, 33, 42, 49, 66, 77, 98, 147, 154, 231, 294, 343, 462, 686, 1029, 1372, 2058, 3234

Prime factors of 72 = 2, 3, 7, 11

{ 2, 3, 7, 11 }

Question 4(vi)

Is $\{ x : x^2 - 7x + 12 = 0 \} = \{ 3, 4 \}$?

Answer

$$x^2 - 7x + 12 = 0$$

$$\Rightarrow x^2 - 3x - 4x + 12 = 0$$

$$\Rightarrow (x^2 - 3x) - (4x - 12) = 0$$

$$\Rightarrow x(x - 3) - 4(x - 3) = 0$$

$$\Rightarrow (x - 3)(x - 4) = 0$$

$$\Rightarrow (x - 3) = 0 \text{ or } (x - 4) = 0$$

$$\Rightarrow x = 3 \text{ or } x = 4$$

Yes, $\{ x : x^2 - 7x + 12 = 0 \} = \{ 3, 4 \}$.

Question 4(vii)

$$\text{Is } \{ x : x^2 - 5x - 6 = 0 \} = \{ 2, 3 \} ?$$

Answer

$$x^2 - 5x - 6 = 0$$

$$\Rightarrow x^2 - 6x + x - 6 = 0$$

$$\Rightarrow (x^2 - 6x) + (x - 6) = 0$$

$$\Rightarrow x(x - 6) + 1(x - 6) = 0$$

$$\Rightarrow (x - 6)(x + 1) = 0$$

$$\Rightarrow (x - 6) = 0 \text{ or } (x + 1) = 0$$

$$\Rightarrow x = 6 \text{ or } x = -1$$

Hence, $\{x : x^2 - 5x - 6 = 0\} \neq \{2, 3\}$.

Question 5(i)

Write the following set in Roster form :

The set of letters in the word 'MEERUT'.

Answer

All letters will be written in the set and there should be no repetition. So all letters are - m, e, r, u and t.

Set = {m, e, r, u, t}

Question 5(ii)

Write the following set in Roster form :

The set of letters in the word 'UNIVERSAL'.

Answer

All letters will be written in the set and there should be no repetition. So all letters are - u, n, i, v, e, r, s, a and l.

Set = {u, n, i, v, e, r, s, a, l}

Question 5(iii)

Write the following set in Roster form :

$$A = \{x : x = y + 3, y \in \mathbb{N} \text{ and } y > 3\}.$$

Answer

$$A = \{x : x = y + 3, y \in \mathbb{N} \text{ and } y > 3\}$$

$$A = \{x : x = 4 + 3, 5 + 3, 6 + 3, \dots\}$$

$$A = \{x : x = 7, 8, 9, \dots\}$$

$$\mathbf{A = \{7, 8, 9, \dots\}}$$

Question 5(iv)

Write the following set in Roster form :

$$B = \{p : p \in \mathbb{W} \text{ and } p^2 < 20\}.$$

Answer

$$B = \{p : p \in \mathbb{W} \text{ and } p^2 < 20\}$$

$$B = \{p : p^2 = 0, 1, 4, 9, 16\}.$$

$$B = \{p : p = 0, 1, 2, 3, 4\}.$$

$$\mathbf{Hence, B = \{0, 1, 2, 3, 4\}.$$

Question 5(v)

Write the following set in Roster form :

$$C = \{x : x \text{ is a composite number and } 5 \leq x \leq 21\}$$

Answer

$$C = \{ x : x \text{ is a composite number and } 5 \leq x \leq 21 \}$$

$$C = \{ x : x = 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21 \}$$

Hence, $C = \{6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21\}$.

Question 6(i)

List the elements of the following set :

$$\{ x : x^2 - 2x - 3 = 0 \}$$

Answer

$$\{ x : x^2 - 2x - 3 = 0 \}$$

$$x^2 - 2x - 3 = 0$$

$$x^2 - 3x + x - 3 = 0$$

$$(x^2 - 3x) + (x - 3) = 0$$

$$x(x - 3) + 1(x - 3) = 0$$

$$(x - 3)(x + 1) = 0$$

$$x - 3 = 0 \text{ or } x + 1 = 0$$

$$x = 3 \text{ or } x = -1$$

Hence, the elements are 3 and -1

Question 6(ii)

List the elements of the following set :

$$\{x : x = 2y + 5 ; y \in \mathbb{N} \text{ and } 2 \leq y < 6\}$$

Answer

$$\{x : x = 2y + 5 ; y \in \mathbb{N} \text{ and } 2 \leq y < 6\}$$

When $y = 2$

$$x = 2 \times 2 + 5 = 4 + 5 = 9$$

When $y = 3$

$$x = 2 \times 3 + 5 = 6 + 5 = 11$$

When $y = 4$

$$x = 2 \times 4 + 5 = 8 + 5 = 13$$

When $y = 5$

$$x = 2 \times 5 + 5 = 10 + 5 = 15$$

Hence, the elements are = 9, 11, 13 and 15.

Question 6(iii)

List the elements of the following set :

$$\{x : x \text{ is a factor of } 24\}$$

Answer

$$\{x : x \text{ is a factor of } 24\}$$

The factors of 24 are 1, 2, 3, 4, 6, 8, 12, 24

Hence, the elements are = 1, 2, 3, 4, 6, 8, 12 and 24.

Question 6(iv)

List the elements of the following set :

$$\{x : x \in \mathbb{Z} \text{ and } x^2 \leq 4\}$$

Answer

$$\{x : x \in \mathbb{Z} \text{ and } x^2 \leq 4\}$$

$$\text{When } x^2 = 4$$

$$x = \sqrt{4} = 2 \text{ or } -2$$

$$\text{When } x^2 = 1$$

$$x = \sqrt{1} = 1 \text{ or } -1$$

$$\text{When } x^2 = 0$$

$$x = \sqrt{0} = 0$$

Hence, the elements are = -2, -1, 0, 1 and 2.

Question 6(v)

List the elements of the following set :

$$\{x : 3x - 2 \leq 10 \text{ and } x \in \mathbb{N}\}$$

Answer

$$\{x : 3x - 2 \leq 10 \text{ and } x \in \mathbb{N}\}$$

$$3x - 2 \leq 10$$

$$3x \leq 10 + 2$$

$$3x \leq 12$$

$$x \leq \frac{12}{3}$$

$$x \leq 4$$

Hence, the elements are = 1, 2, 3, and 4.

Question 6(vi)

List the elements of the following set :

$$\{x : 4 - 2x > -6, x \in \mathbb{Z}\}$$

Answer

$$\{x : 4 - 2x > -6, x \in \mathbb{Z}\}$$

$$4 - 2x > -6$$

$$\Rightarrow -4 + 4 - 2x > -6 - 4$$

$$\Rightarrow -2x > -10$$

$$\Rightarrow 2x < 10$$

$$\Rightarrow x < \frac{10}{2}$$

$$\Rightarrow x < 5$$

Hence, the elements are = 4, 3, 2, 1, 0, -1, -2,

Question 7

State which of the following sets are finite and which are infinite :

(i) $A = \{x : x \in \mathbb{Z} \text{ and } x < 10\}$

(ii) $B = \{x : x \in \mathbb{W} \text{ and } 5x - 3 \leq 20\}$

(iii) $P = \{y : y = 3x - 2, x \in \mathbb{N} \text{ and } x > 5\}$

(iv) $M = \left\{x : x = \frac{3}{n}; n \in \mathbb{W} \text{ and } 6 < n \leq 15\right\}$

Answer

(i) Infinite.

Reason

The set contains all negative numbers , 0 and natural numbers till 10:

$$A = \{\dots, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

(ii) Finite

Reason

$$5x - 3 \leq 20$$

$$\Rightarrow 5x \leq 20 + 3$$

$$\Rightarrow 5x \leq 23$$

$$\Rightarrow x \leq \frac{23}{5}$$

$$\Rightarrow x \leq 4.6$$

$$B = \{0, 1, 2, 3, 4\}$$

(iii) Infinite

Reason

$$y = 3x - 2$$

When $x = 6$

$$y = 3 \times 6 - 2 = 18 - 2 = 16$$

When $x = 7$

$$y = 3 \times 7 - 2 = 21 - 2 = 19$$

When $x = 8$

$$y = 3 \times 8 - 2 = 24 - 2 = 22$$

When $x = 9$

$$y = 3 \times 9 - 2 = 27 - 2 = 25$$

$$P = \{16, 19, 22, 25, \dots\}$$

(iv) Finite

Reason

When $n = 7$

$$x = \frac{3}{7}$$

When $n = 8$

$$x = \frac{3}{8}$$

When $n = 9$

$$x = \frac{3}{9}$$

When $n = 10$

$$x = \frac{3}{10}$$

When $n = 11$

$$x = \frac{3}{11}$$

When $n = 12$

$$x = \frac{3}{12}$$

When $n = 13$

$$x = \frac{3}{13}$$

When $n = 14$

$$x = \frac{3}{14}$$

When $n = 15$

$$x = \frac{3}{15}$$

$$M = \left\{ \frac{3}{7}, \frac{3}{8}, \frac{3}{9}, \frac{3}{10}, \frac{3}{11}, \frac{3}{12}, \frac{3}{13}, \frac{3}{14}, \frac{3}{15} \right\}$$

Reducing the fractions:

$$M = \left\{ \frac{3}{7}, \frac{3}{8}, \frac{1}{3}, \frac{3}{10}, \frac{3}{11}, \frac{1}{4}, \frac{3}{13}, \frac{3}{14}, \frac{1}{5} \right\}$$

Question 8

Find which of the following sets are singleton sets :

(i) The set of points of intersection of two non-parallel straight lines on the same plane.

(ii) $A = \{x : 7x - 3 = 11\}$

(iii) $B = \{y : 2y + 1 < 3 \text{ and } y \in W\}$

Answer

(i) There will be always one and only point where two non-parallel straight lines meet on the same plane.

Hence, this is a singleton set.

(ii) $7x - 3 = 11$

$$7x = 11 + 3$$

$$7x = 14$$

$$x = \frac{14}{7}$$

$$x = 2$$

$A = \{2\}$. Hence, A is a singleton set.

$$(iii) 2y + 1 < 3$$

$$2y < 3 - 1$$

$$2y < 2$$

$$y < \frac{2}{2}$$

$$y < 1$$

B = {0}. Hence, B is a singleton set.

Question 9

Find which of the following sets are empty :

(i) The set of points of intersection of two parallel lines.

(ii) $A = \{x : x \in \mathbb{N} \text{ and } 5 < x \leq 6\}$.

(iii) $B = \{x : x^2 + 4 = 0 \text{ and } x \in \mathbb{N}\}$.

(iv) $C = \{\text{even numbers between 6 and 10}\}$.

(v) $D = \{\text{prime numbers between 7 and 11}\}$.

Answer

(i) Parallel lines don't intersect, hence their points of intersection is an **empty set**.

(ii) $A = \{6\}$

$A \neq \{\}$

$$(iii) x^2 + 4 = 0$$

$$x^2 = -4$$

$$x = \sqrt{-4}$$

$$B = \{\sqrt{-4}\}$$

As square root of negative number is undefined,

$$\text{Hence, } B = \{\}$$

$$(iv) C = \{6, 8, 10\}$$

$$C \neq \{\}$$

(v) No prime number between 7 and 11.

$$D = \{\}$$

Question 10(i)

Are the sets $A = \{4, 5, 6\}$ and

$B = \{x : x^2 - 5x - 6 = 0\}$ disjoint ?

Answer

$$A = \{4, 5, 6\}$$

$$B = \{x : x^2 - 5x - 6 = 0\}$$

$$\Rightarrow B = \{x : x^2 - 6x + x - 6 = 0\}$$

$$\Rightarrow B = \{x : x(x - 6) + 1(x - 6) = 0\}$$

$$\Rightarrow B = \{x : (x - 6)(x + 1) = 0\}$$

$$\Rightarrow B = \{x : (x - 6) = 0 \text{ or } (x + 1) = 0\}$$

$$\Rightarrow B = \{x : x = 6 \text{ or } x = -1\}$$

$$\Rightarrow B = \{6, -1\}$$

As 6 is a common element in A and B,

Hence, A and B are not disjoint sets.

Question 10(ii)

Are the sets $A = \{b, c, d, e\}$ and

$B = \{x : x \text{ is a letter in the word 'MASTER'}\}$ joint ?

Answer

$$A = \{b, c, d, e\}$$

$$B = \{x : x \text{ is a letter in the word 'MASTER'}\}$$

$$B = \{m, a, s, t, e, r\}$$

As e is a common element in A and B,

Hence, A and B are joint sets.

Question 11

State whether the following pairs of sets are equivalent or not :

(i) $A = \{x : x \in \mathbb{N} \text{ and } 11 \geq 2x - 1\}$ and

$$B = \{y : y \in \mathbb{W} \text{ and } 3 \leq y \leq 9\}.$$

(ii) Set of integers and set of natural numbers.

(iii) Set of whole numbers and set of multiples of 3.

(iv) $P = \{5, 6, 7, 8\}$ and $M = \{x : x \in W \text{ and } x \leq 4\}$.

Answer

(i) Not equivalent

Reason

$$A = \{x : x \in N \text{ and } 11 \geq 2x - 1\}$$

$$A = \{11 \geq 2x - 1\}$$

$$A = \{11 + 1 \geq 2x\}$$

$$A = \{12 \geq 2x\}$$

$$A = \{x \leq \frac{12}{2}\}$$

$$A = \{x \leq 6\}$$

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$B = \{y : y \in W \text{ and } 3 \leq y \leq 9\}$$

$$B = \{3, 4, 5, 6, 7, 8, 9\}$$

There are 6 elements in set A and 7 in set B.

(ii) Not equivalent

Reason

Set of integers contain all negative numbers , 0 and positive numbers.

$$A = \{....., -2, -1, 0, 1, 2,.....\}$$

Set of natural numbers contain all positive numbers.

$$B = \{1, 2, 3, 4,\}$$

(iii) Not equivalent

Reason

Set of whole numbers contains all natural numbers and 0.

$$X = \{0, 1, 2, 3, 4,.....\}$$

Set of multiples of 3 contains all the numbers which are multiple of 3.

$$Y = \{3, 6, 9, 12, 15, 18,.....\}$$

Hence, they contain different number of questions.

(iv) Not equivalent

Reason

$$P = \{5, 6, 7, 8\}$$

$$M = \{x : x \in W \text{ and } x \leq 4\}$$

$$M = \{0, 1, 2, 3, 4\}$$

As P contains 4 elements whereas M contains 5 elements.

Question 12

State whether the following pairs of sets are equal or not :

(i) $A = \{2, 4, 6, 8\}$ and $B = \{2n : n \in \mathbb{N} \text{ and } n < 5\}$

(ii) $M = \{x : x \in \mathbb{W} \text{ and } x + 3 < 8\}$ and $N = \{y : y = 2n - 1, n \in \mathbb{N} \text{ and } n < 5\}$

(iii) $E = \{x : x^2 + 8x - 9 = 0\}$ and $F = \{1, -9\}$

(iv) $A = \{x : x \in \mathbb{N}, x < 3\}$ and

$B = \{y : y^2 - 3y + 2 = 0\}$

Answer

(i) Equal

Reason

$$A = \{2, 4, 6, 8\}$$

$$B = \{2n : n \in \mathbb{N} \text{ and } n < 5\}$$

$$\text{When } n = 1$$

$$2n = 2 \times 1 = 2$$

$$\text{When } n = 2$$

$$2n = 2 \times 2 = 4$$

$$\text{When } n = 3$$

$$2n = 2 \times 3 = 6$$

$$\text{When } n = 4$$

$$2n = 2 \times 4 = 8$$

$$B = \{2, 4, 6, 8\}$$

Hence, $A = B$

(ii) Not equal

Reason

$$M = \{x : x \in W \text{ and } x + 3 < 8\}$$

$$x + 3 < 8$$

$$\Rightarrow x < 8 - 3$$

$$\Rightarrow x < 5$$

$$M = \{0, 1, 2, 3, 4\}$$

$$N = \{y : y = 2n - 1, n \in N \text{ and } n < 5\}$$

When $n = 1$

$$y = 2 \times 1 - 1 = 2 - 1 = 1$$

When $n = 2$

$$y = 2 \times 2 - 1 = 4 - 1 = 3$$

When $n = 3$

$$y = 2 \times 3 - 1 = 6 - 1 = 5$$

When $n = 4$

$$y = 2 \times 4 - 1 = 8 - 1 = 7$$

$$N = \{1, 3, 5, 7\}$$

Hence, $M \neq N$

(iii) Equal

Reason

$$E = \{x : x^2 + 8x - 9 = 0\}$$

$$x^2 + 8x - 9 = 0$$

$$\Rightarrow x^2 + 9x - x - 9 = 0$$

$$\Rightarrow x(x + 9) - 1(x + 9) = 0$$

$$\Rightarrow (x + 9)(x - 1) = 0$$

$$\Rightarrow (x + 9) = 0 \text{ or } (x - 1) = 0$$

$$\Rightarrow x = -9 \text{ or } 1$$

$$E = \{1, -9\}$$

$$F = \{1, -9\}$$

Hence, $E = F$

(iv) Equal

Reason

$$A = \{x : x \in \mathbb{N}, x < 3\}$$

$$A = \{1, 2\}$$

$$B = \{y : y^2 - 3y + 2 = 0\}$$

$$y^2 - 3y + 2 = 0$$

$$\Rightarrow y^2 - 2y - y + 2 = 0$$

$$\Rightarrow y(y - 2) - 1(y - 2) = 0$$

$$\Rightarrow (y - 2)(y - 1) = 0$$

$$\Rightarrow (y - 2) = 0 \text{ or } (y - 1) = 0$$

$$\Rightarrow y = 2 \text{ or } y = 1$$

$$\Rightarrow y = \{1, 2\}$$

Hence, $A = B$

Question 13

State whether each of the following sets is a finite set or an infinite set :

(i) The set of multiples of 8.

(ii) The set of integers less than 10.

(iii) The set of whole numbers less than 12.

(iv) $\{x : x = 3n - 2, n \in W, n \leq 8\}$

(v) $\{x : x = 3n - 2, n \in Z, n \leq 8\}$

(vi) $\left\{x : x = \frac{n - 2}{n + 1}, n \in W\right\}$

Answer

(i) Infinite

Reason

There are infinite multiple of 8.

$$A = \{8, 16, 24, 32, \dots\}$$

(ii) Infinite

Reason

Integers less than 10 includes all negative numbers, 0 and positive number till 9.

$$B = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

(iii) Finite

Reason

Whole numbers less than 12 include all the numbers between 0 to 11

$$C = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

(iv) Finite

Reason

$$x = 3n - 2$$

When $n = 0$

$$x = 3 \times 0 - 2 = 0 - 2 = -2$$

When $n = 1$

$$x = 3 \times 1 - 2 = 3 - 2 = 1$$

When $n = 2$

$$x = 3 \times 2 - 2 = 6 - 2 = 4$$

When $n = 3$

$$x = 3 \times 3 - 2 = 9 - 2 = 7$$

When $n = 4$

$$x = 3 \times 4 - 2 = 12 - 2 = 10$$

When $n = 5$

$$x = 3 \times 5 - 2 = 15 - 2 = 13$$

When $n = 6$

$$x = 3 \times 6 - 2 = 18 - 2 = 16$$

When $n = 7$

$$x = 3 \times 7 - 2 = 21 - 2 = 19$$

When $n = 8$

$$x = 3 \times 8 - 2 = 24 - 2 = 22$$

$$D = \{-2, 1, 4, 7, 10, 13, 16, 19, 22\}$$

(v) Infinite

Reason

Integers less than equal to 8 include all negative numbers, 0 and positive numbers till 8.

$$D = \{\dots\dots\dots, -8, -5, -2, 1, 4, 7, 10\}$$

(vi) Infinite

Reason

$$\left\{ x : x = \frac{n-2}{n+1}, n \in W \right\}$$

When $n = 0$

$$x = \frac{0-2}{0+1} = \frac{-2}{1} = -1$$

When $n = 1$

$$x = \frac{1-2}{1+1} = \frac{-1}{2}$$

When $n = 2$

$$x = \frac{2-2}{2+1} = \frac{0}{3} = 0$$

When $n = 3$

$$x = \frac{3-2}{3+1} = \frac{1}{4}$$

Similarly $n = 4, 5, 6, \dots\dots\dots$ and hence we get the value of x for every value of n .

Question 14

Answer whether the following statements are **true** or **false**. Give reasons.

- (i) The set of even natural numbers less than 21 and the set of odd natural numbers less than 21 are equivalent sets.
- (ii) If $E = \{\text{factors of } 16\}$ and $F = \{\text{factors of } 20\}$, then $E = F$.
- (iii) The set $A = \{\text{integers less than } 20\}$ is a finite set.
- (iv) If $A = \{x : x \text{ is an even prime number}\}$, then set A is empty.
- (v) The set of odd prime numbers is the empty set.
- (vi) The set of squares of integers and the set of whole numbers are equal sets.

Answer

(i) True

Reason

The set of even natural numbers less than 21

$$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

The set of odd natural numbers less than 21

$$B = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$$

A and B both contains 10 elements. Hence they are equivalent sets.

(ii) False

Reason

$$E = \{\text{factors of } 16\}$$

$$\text{Factors of } 16 = 1, 2, 4, 8, 16$$

$$E = \{1, 2, 4, 8, 16\}$$

$$F = \{\text{factors of } 20\}$$

$$\text{Factors of } 20 = 1, 2, 4, 5, 10, 20$$

$$F = \{1, 2, 4, 5, 10, 20\}$$

Hence, $E \neq F$

(iii) False

Reason

$$A = \{\text{integers less than } 20\}$$

A contains all negative numbers, 0, positive numbers less than 20.

$$A = \{\dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19\}$$

Hence, A is infinite set.

(iv) False

Reason

$$A = \{x : x \text{ is an even prime number}\}$$

$$A = \{2\}$$

Hence, A is not an empty set.

(v) False

Reason

Odd prime numbers = 3, 5, 7, 11, 17,

$$A = \{3, 5, 7, 11, 17, \dots\}$$

Hence, it is not an empty set.

(vi) False

Reason

The set of squares of integers

$$M = \{1^2, 2^2, 3^2, 4^2, \dots\}$$

$$M = \{1, 4, 9, 16, \dots\}$$

The set of whole numbers:

$$N = \{0, 1, 2, 3, 4, \dots\}$$

$$M \neq N$$

Exercise 6(B)

Question 1(i)

A set P has 3 elements. The number of proper subsets of set B is :

1. $3 \times 2 = 6$

2. 8

3. 9

4. 7

Answer

If a set has n elements, the number of proper subsets = $2^n - 1$

$$n = 3$$

$$\text{Number of proper subsets} = 2^3 - 1$$

$$= 8 - 1$$

$$= 7$$

Hence, option 4 is the correct option.

Question 1(ii)

For sets A and B, where $A = \{2, 4, 6\}$ and $B = \{1, 3, 5, 7\}$, $A \cap B$ is :

1. ϕ
2. $\{0\}$
3. $\{1, 2, 3, 4, 5, 6, 7\}$
4. none of these

Answer

$A = \{2, 4, 6\}$ and $B = \{1, 3, 5, 7\}$

No element is common between A and B. Hence,

$$A \cap B = \{2, 4, 6\} \cap \{1, 3, 5, 7\}$$

$$A \cap B = \{\} = \phi$$

Hence, option 1 is the correct option.

Question 1(iii)

If set $A = \{4, 6, 8\}$ and set $B = \{0\}$, then $A \cup B$ is :

1. $\{4, 6, 8\}$
2. $\{4, 6, 8, 0\}$
3. $\{\}$

4. none of these

Answer

set A = {4, 6, 8} and set B = {0}

Elements in both set,

$$A \cup B = \{4, 6, 8\} \cup \{0\}$$

$$A \cup B = \{4, 6, 8, 0\}$$

Hence, option 2 is the correct option.

Question 1(iv)

If set A = students in class 8 of a particular school and set B = students of this school, then :

1. $A = B$

2. $n(A) = n(B)$

3. $A \subset B$

4. $B \subseteq A$

Answer

Set A = students in class 8 of a particular school

Set B = students of this school

All students in class 8 will be the students of the school also.

Hence, A is proper subset of B i.e., $A \subset B$

Hence, option 3 is the correct option.

Question 1(v)

If universal set $\xi = \{x : x \in W, x < 5\}$ and set $A = \{1, 3\}$, then complement of set A is equal to :

1. $\{0, 2, 4, 5\}$

2. $\{0, 2, 4\}$

3. ϕ

4. $\{2, 4\}$

Answer

Universal set $\xi = \{x : x \in W, x < 5\}$

Universal set $\xi = \{0, 1, 2, 3, 4\}$

Set $A = \{1, 3\}$

Complement of A means all the elements of universal set which are not in A.

$A' = \{0, 2, 4\}$

Hence, option 2 is the correct option.

Question 1(vi)

If universal set = N, the set of natural numbers, set $A = \{\text{multiples of 3 less than or equal to 20}\}$

and

Set $B = \{\text{multiples of 4 less than or equal to 20}\}$, then $A - B$ is equal to :

1. $\{3, 6, 9, 15, 18\}$

2. $\{4, 8, 16, 20\}$

3. $\{0\}$

4. $\{12\}$

Answer

Universal set = N , the set of natural numbers,

$= \{1, 2, 3, 4, \dots\}$

Set $A = \{\text{multiples of 3 less than or equal to 20}\}$

$= \{3, 6, 9, 12, 15, 18\}$

Set $B = \{\text{multiples of 4 less than or equal to 20}\}$

$= \{4, 8, 12, 16\}$

$A - B$ - all the elements of A except the common elements of A and B

$= \{3, 6, 9, 15, 18\}$

Hence, option 1 is the correct option.

Question 2(i)

Find all the subsets of $A = \{5, 7\}$.

Answer

If a set has n elements, the number of subsets $= 2^n$.

Here the set has 2 elements, the number of subsets = $2^2 = 2 \times 2 = 4$.

Subsets of A are ϕ , {5}, {7}, {5, 7}.

Question 2(ii)

Find all the subsets of $B = \{a, b, c\}$.

Answer

If a set has n elements, the number of subsets = 2^n .

Here the set has 3 elements, the number of subsets = $2^3 = 2 \times 2 \times 2 = 8$.

Subsets of B are ϕ , {a}, {b}, {c}, {a, b}, {b, c}, {a, c}, {a, b, c}.

Question 2(iii)

Find all the subsets of $C = \{x : x \in W, x \leq 2\}$.

Answer

$C = \{0, 1, 2\}$

If a set has n elements, the number of subsets = 2^n .

Here, the set has 3 elements, the number of subsets = $2^3 = 2 \times 2 \times 2 = 8$.

Subsets of B are ϕ , {0}, {1}, {2}, {0, 1}, {0, 2}, {1, 2}, {0, 1, 2}.

Question 2(iv)

Find all the subsets of $\{p : p \text{ is a letter in the word 'poor'}\}$.

Answer

$$C = \{p, o, r\}$$

If a set has n elements, the number of subsets = 2^n .

Here the set has 3 elements, the number of subsets = $2^3 = 2 \times 2 \times 2 = 8$.

Subsets of B are ϕ , $\{p\}$, $\{o\}$, $\{r\}$, $\{p, o\}$, $\{o, r\}$, $\{r, p\}$, $\{p, o, r\}$.

Question 3

If C is the set of letters in the word 'cooler', find:

(i) set C

(ii) $n(C)$

(iii) number of its subsets

(iv) number of its proper subsets

Answer

(i) Set C contains letters of word 'cooler' — c, o, l, e, r.

$$C = \{c, o, l, e, r\}$$

(ii) $n(C)$ is the cardinal number of set C.

$$n(C) = 5$$

(iii) Here the set has 5 elements, the number of subsets = $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Number of subsets of B are 32.

(iv) Here the set has 5 elements, the number of subsets = $2^5 - 1 = 2 \times 2 \times 2 \times 2 \times 2 - 1 = 32 - 1 = 31$.

Number of proper subsets of B are 31.

Question 4

If $T = \{x : x \text{ is a letter in the word 'TEETH'}\}$, find all its subsets.

Answer

Set T contains all the letters of 'TEETH'.

$$T = \{t, e, h\}$$

If a set has n elements, the number of subsets = 2^n .

Here the set has 3 elements, the number of subsets = $2^3 = 2 \times 2 \times 2 = 8$.

Subsets of T are ϕ , {t}, {e}, {h}, {t, e}, {e, h}, {h, t}, {t, e, h}.

Question 5

Given the universal set = $\{-7, -3, -1, 0, 5, 6, 8, 9\}$, find :

(i) $A = \{x : x < 2\}$

(ii) $B = \{x : -4 < x < 6\}$

Answer

(i) A contains elements less than 2 i.e, -7, -3, -1 and 0.

$$A = \{-7, -3, -1, 0\}$$

(ii) B contains all the elements between -4 and 6 i.e, -3, -1, 0 and 5.

$$B = \{-3, -1, 0, 5\}$$

Question 6

Given the universal set = $\{x : x \in \mathbb{N} \text{ and } x < 20\}$, find :

(i) $A = \{x : x = 3p ; p \in \mathbb{N}\}$

(ii) $B = \{y : y = 2n + 3, n \in \mathbb{N}\}$

(iii) $C = \{x : x \text{ is divisible by } 4\}$

Answer

(i) $x = 3p$

When $p = 1$

$$x = 3p = 3 \times 1 = 3$$

When $p = 2$

$$x = 3p = 3 \times 2 = 6$$

When $p = 3$

$$x = 3p = 3 \times 3 = 9$$

When $p = 4$

$$x = 3p = 3 \times 4 = 12$$

When $p = 5$

$$x = 3p = 3 \times 5 = 15$$

When $p = 6$

$$x = 3p = 3 \times 6 = 18$$

$$\mathbf{A = \{3, 6, 9, 12, 15, 18\}}$$

$$(ii) y = 2n + 3$$

When $n = 1$

$$y = 2n + 3 = 2 \times 1 + 3 = 5$$

When $n = 2$

$$y = 2n + 3 = 2 \times 2 + 3 = 7$$

When $n = 3$

$$y = 2n + 3 = 2 \times 3 + 3 = 9$$

When $n = 4$

$$y = 2n + 3 = 2 \times 4 + 3 = 11$$

When $n = 5$

$$y = 2n + 3 = 2 \times 5 + 3 = 13$$

When $n = 6$

$$y = 2n + 3 = 2 \times 6 + 3 = 15$$

When $n = 7$

$$y = 2n + 3 = 2 \times 7 + 3 = 17$$

When $n = 8$

$$y = 2n + 3 = 2 \times 8 + 3 = 19$$

$$\mathbf{B = \{5, 7, 9, 11, 13, 15, 17, 19\}}$$

(iii) x is divisible by 4.

$$x = 4, 8, 12, 16$$

$$\mathbf{C = \{4, 8, 12, 16\}}$$

Question 7

Find the proper subsets of $\{x : x^2 - 9x - 10 = 0\}$.

Answer

$$x^2 - 9x - 10 = 0$$

$$\Rightarrow x^2 - 10x + x - 10 = 0$$

$$\Rightarrow x(x - 10) + 1(x - 10) = 0$$

$$\Rightarrow (x - 10)(x + 1) = 0$$

$$\Rightarrow (x - 10) = 0 \text{ or } (x + 1) = 0$$

$$\Rightarrow x = 10 \text{ or } x = -1$$

$$\Rightarrow A = \{10, -1\}$$

If a set has n elements, the number of proper subsets = $2^n - 1$.

Here the set has 2 elements, the number of proper subsets = $2^2 - 1 = 2 \times 2 - 1 = 3$.

Proper subsets of A are ϕ , {10}, {-1}.

Question 8

Given, $A = \{\text{Triangles}\}$, $B = \{\text{Isosceles triangles}\}$, $C = \{\text{Equilateral triangles}\}$.

State whether the following are **true** or **false**. Give reasons.

(i) $A \subset B$

(ii) $B \subseteq A$

(iii) $C \subseteq B$

(iv) $B \subset A$

(v) $C \subset A$

(vi) $C \subseteq B \subseteq A$.

Answer

A is a set which contains all triangles.

B is a set which contains all isosceles triangles.

C is a set which contains all equilateral triangles.

(i) False

Reason

As we know each triangle cannot be isosceles triangle.

Hence, A is not a proper subset of B.

(ii) True

Reason

As all isosceles triangles are type of triangles.

Hence, B is a subset of A.

(iii) True

Reason

As all equilateral triangles are isosceles triangles also.

Hence, C is a subset of B.

(iv) True

Reason

As all isosceles triangles are triangle also.

Hence, B is a proper subset of A.

(v) True

Reason

As all equilateral triangles are triangle also.

Hence, C is a proper subset of A.

(vi) True

Reason

As all equilateral triangles are isosceles triangle and all isosceles triangles are triangle also.

Hence, C is a subset of B and B is a subset of A.

Question 9

Given, $A = \{\text{Quadrilaterals}\}$, $B = \{\text{Rectangles}\}$, $C = \{\text{Squares}\}$ and $D = \{\text{Rhombuses}\}$. State, giving reasons, whether the following are **true** or **false**.

(i) $B \subset C$

(ii) $D \subset B$

(iii) $C \subseteq B \subseteq A$

(iv) $D \subset A$

(v) $B \supseteq C$

(vi) $A \supseteq B \supseteq D$

Answer

A is set of all quadrilaterals.

B is set of all rectangles.

C is set of all squares.

D is set of all rhombuses.

(i) False

Reason

As all rectangles are not squares.

Hence, B is a not proper subset of C.

(ii) False

Reason

As all rhombuses are not rectangles.

Hence, D is a not proper subset of B.

(iii) True

Reason

As all squares are rectangles and all rectangles are quadrilateral.

Hence, C is a subset of B and B is a subset of A.

(iv) True

Reason

As all rhombuses are quadrilateral.

Hence, D is a proper subset of A.

(v) True

Reason

As all squares are rectangles.

Hence, C is a subset of B.

(vi) False

Reason

As all rectangles are quadrilateral but all rhombuses are not rectangles.

Hence, B is a subset of A and D is not a subset of B.

Question 10

Given, universal set = $\{x : x \in \mathbb{N}, 10 \leq x \leq 35\}$, $A = \{x \in \mathbb{N} : x \leq 16\}$ and $B = \{x : x > 29\}$. Find:

(i) A'

(ii) B'

Answer

Universal set = $\{x : x \in \mathbb{N}, 10 \leq x \leq 35\}$

Universal set = $\{10, 11, 12, 13, 14, 15, 16, \dots, 34, 35\}$

$A = \{x \in \mathbb{N} : x \leq 16\}$

$B = \{x : x > 29\}$

(i) $A = \{10, 11, 12, 13, 14, 15, 16\}$

A' - contains all elements in universal set which are not in set A.

$A' = \{17, 18, 19, 20, 21, \dots, 34, 35\}$

$A' = \{x : x \in \mathbb{N}, 17 \leq x \leq 35\}$

$$(ii) B = \{30, 31, 32, 33, 34, 35\}$$

B' - contains all elements in universal set which are not in set B .

$$B' = \{10, 11, 12, 13, 14, 15, 16, \dots, 28, 29\}$$

$$B' = \{x : x \in N, 10 \leq x \leq 29\}$$

Question 11

Given, universal set = $\{x \in Z : -6 < x \leq 6\}$, $N = \{n : n \text{ is a non-negative number}\}$ and $P = \{x : x \text{ is a non-positive number}\}$. Find :

(i) N'

(ii) P'

Answer

$$\text{Universal set} = \{x \in Z : -6 < x \leq 6\}$$

$$\text{Universal set} = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$$

$$N = \{n : n \text{ is a non-negative number}\}$$

$$P = \{x : x \text{ is a non-positive number}\}$$

$$(i) N = \{0, 1, 2, 3, 4, 5, 6\}$$

N' - contains all elements in universal set which are not in set N .

$$N' = \{-5, -4, -3, -2, -1\}$$

$$(ii) P = \{-5, -4, -3, -2, -1, 0\}$$

P' - contains all elements in universal set which are not in set P .

$$P' = \{1, 2, 3, 4, 5, 6\}$$

Question 12

Let $M = \{\text{letters of the word REAL}\}$ and $N = \{\text{letters of the word LARE}\}$. Write sets M and N in roster form and then state whether;

(i) $M \subseteq N$ is true.

(ii) $N \subseteq M$ is true.

(iii) $M = N$ is true.

Answer

$M = \{\text{letters of the word REAL}\}$

$M = \{R, E, A, L\}$

$N = \{\text{letters of the word LARE}\}$

$N = \{L, A, R, E\}$

(i) True

Reason

Every elements in set M is also in set N .

(ii) True

Reason

Every elements in set N is also in set M .

(iii) True

Reason

Both sets contain the same elements.

Question 13

Given $A = \{x : x \in \mathbb{N} \text{ and } 3 < x \leq 6\}$ and $B = \{x : x \in \mathbb{W} \text{ and } x < 4\}$. Find :

(i) sets A and B in roster form

(ii) $A \cup B$

(iii) $A \cap B$

(iv) $A - B$

(v) $B - A$

Answer

$$A = \{x : x \in \mathbb{N} \text{ and } 3 < x \leq 6\}$$

$$B = \{x : x \in \mathbb{W} \text{ and } x < 4\}$$

$$(i) A = \{4, 5, 6\}$$

$$B = \{0, 1, 2, 3\}$$

(ii) $A \cup B$ - contains all the elements in both sets A and B.

$$A \cup B = \{4, 5, 6\} \cup \{0, 1, 2, 3\}$$

$$A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$$

(iii) $A \cap B$ - contains all the common elements in both sets A and B.

$$A \cap B = \{4, 5, 6\} \cap \{0, 1, 2, 3\}$$

$$A \cap B = \phi$$

(iv) $A - B$ - includes all elements in set A but not in set B.

$$A - B = \{4, 5, 6\} - \{0, 1, 2, 3\}$$

$$A - B = \{4, 5, 6\}$$

(v) $B - A$ - includes all elements in set B but not in set A.

$$B - A = \{0, 1, 2, 3\} - \{4, 5, 6\}$$

$$B - A = \{0, 1, 2, 3\}$$

Question 14

If $P = \{x : x \in W \text{ and } 4 \leq x \leq 8\}$ and $Q = \{x : x \in N \text{ and } x < 6\}$. Find

(i) $P \cup Q$ and $P \cap Q$.

(ii) Is $(P \cup Q) \supset (P \cap Q)$?

Answer

$$P = \{x : x \in W \text{ and } 4 \leq x \leq 8\}$$

$$P = \{4, 5, 6, 7, 8\}$$

$$Q = \{x : x \in N \text{ and } x < 6\}$$

$$Q = \{1, 2, 3, 4, 5\}$$

(i) $P \cup Q$ - contains all the elements in both sets P and Q.

$$P \cup Q = \{4, 5, 6, 7, 8\} \cup \{1, 2, 3, 4, 5\}$$

$$P \cup Q = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$P \cap Q$ - contains all the common elements in both sets P and Q.

$$P \cap Q = \{4, 5, 6, 7, 8\} \cap \{1, 2, 3, 4, 5\}$$

$$P \cap Q = \{4, 5\}$$

(ii) From (i) we can clearly see that every elements in $P \cap Q$ is also in $P \cup Q$.

Hence, $(P \cup Q) \supset (P \cap Q)$ is correct.

Question 15

If $A = \{5, 6, 7, 8, 9\}$, $B = \{x : 3 < x < 8 \text{ and } x \in W\}$ and $C = \{x : x \leq 5 \text{ and } x \in N\}$.

Find :

(i) $A \cup B$ and $(A \cup B) \cup C$

(ii) $B \cup C$ and $A \cup (B \cup C)$

(iii) $A \cap B$ and $(A \cap B) \cap C$

(iv) $B \cap C$ and $A \cap (B \cap C)$

Is $(A \cup B) \cup C = A \cup (B \cup C)$?

Is $(A \cap B) \cap C = A \cap (B \cap C)$?

Answer

$$A = \{5, 6, 7, 8, 9\}$$

$$B = \{x : 3 < x < 8 \text{ and } x \in W\}$$

$$B = \{4, 5, 6, 7\}$$

$$C = \{x : x \leq 5 \text{ and } x \in \mathbb{N}\}$$

$$C = \{1, 2, 3, 4, 5\}$$

(i) $A \cup B$ - contains all elements in both set A and B.

$$A \cup B = \{5, 6, 7, 8, 9\} \cup \{4, 5, 6, 7\}$$

$$\mathbf{A \cup B = \{4, 5, 6, 7, 8, 9\}}$$

$(A \cup B) \cup C$ - contains all elements in set A, B and C.

$$(A \cup B) \cup C = \{4, 5, 6, 7, 8, 9\} \cup \{1, 2, 3, 4, 5\}$$

$$\mathbf{(A \cup B) \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}}$$

(ii) $B \cup C$ - contains all elements in both set B and C.

$$B \cup C = \{4, 5, 6, 7\} \cup \{1, 2, 3, 4, 5\}$$

$$\mathbf{B \cup C = \{1, 2, 3, 4, 5, 6, 7\}}$$

$A \cup (B \cup C)$ - contains all elements in set A, B and C.

$$A \cup (B \cup C) = \{5, 6, 7, 8, 9\} \cup \{1, 2, 3, 4, 5, 6, 7\}$$

$$\mathbf{A \cup (B \cup C) = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}}$$

(iii) $A \cap B$ - contains common elements in both set A and B.

$$A \cap B = \{5, 6, 7, 8, 9\} \cap \{4, 5, 6, 7\}$$

$$\mathbf{A \cap B = \{5, 6, 7\}}$$

$(A \cap B) \cap C$ - contains common elements in set A, B and C.

$$(A \cap B) \cap C = \{5, 6, 7\} \cap \{1, 2, 3, 4, 5\}$$

$$(A \cap B) \cap C = \{5\}$$

(iv) $B \cap C$ - contains common elements in both set B and C.

$$B \cap C = \{4, 5, 6, 7\} \cap \{1, 2, 3, 4, 5\}$$

$$B \cap C = \{4, 5\}$$

$A \cap (B \cap C)$ - contains common elements in set A, B and C.

$$A \cap (B \cap C) = \{5, 6, 7, 8, 9\} \cap \{4, 5\}$$

$$A \cap (B \cap C) = \{5\}$$

As we have seen in (i) and (ii), all elements in $(A \cup B) \cup C$ and $A \cup (B \cup C)$ are same.

$$(A \cup B) \cup C = A \cup (B \cup C)$$

As we have seen in (iii) and (iv), all elements in $(A \cap B) \cap C$ and $A \cap (B \cap C)$ are same.

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Question 16

Given $A = \{0, 1, 2, 4, 5\}$, $B = \{0, 2, 4, 6, 8\}$ and $C = \{0, 3, 6, 9\}$. Show that :

(i) $A \cup (B \cup C) = (A \cup B) \cup C$, i.e., the union of sets is associative.

(ii) $A \cap (B \cap C) = (A \cap B) \cap C$, i.e., the intersection of sets is associative.

Answer

$$A = \{0, 1, 2, 4, 5\}$$

$$B = \{0, 2, 4, 6, 8\}$$

$$C = \{0, 3, 6, 9\}$$

(i) $B \cup C$ - contains all the elements in set B and C.

$$B \cup C = \{0, 2, 4, 6, 8\} \cup \{0, 3, 6, 9\}$$

$$B \cup C = \{0, 2, 3, 4, 6, 8, 9\}$$

$A \cup (B \cup C)$ - contains all the elements in set A, B and C.

$$A \cup (B \cup C) = \{0, 1, 2, 4, 5\} \cup \{0, 2, 3, 4, 6, 8, 9\}$$

$$A \cup (B \cup C) = \{0, 1, 2, 3, 4, 5, 6, 8, 9\} \dots\dots\dots(1)$$

$A \cup B$ - contains all the elements in set A and B.

$$A \cup B = \{0, 1, 2, 4, 5\} \cup \{0, 2, 4, 6, 8\}$$

$$A \cup B = \{0, 1, 2, 4, 5, 6, 8\}$$

$(A \cup B) \cup C$ - contains all the elements in set A, B and C.

$$(A \cup B) \cup C = \{0, 1, 2, 4, 5, 6, 8\} \cup \{0, 3, 6, 9\}$$

$$(A \cup B) \cup C = \{0, 1, 2, 3, 4, 5, 6, 8, 9\} \dots\dots\dots(2)$$

Hence from (1) and (2), all the elements of $A \cup (B \cup C)$ and $(A \cup B) \cup C$ are same.

$A \cup (B \cup C) = (A \cup B) \cup C$, i.e. the union of sets is associative.

(ii) $B \cap C$ - contains all the common elements in set B and C.

$$B \cap C = \{0, 2, 4, 6, 8\} \cap \{0, 3, 6, 9\}$$

$$B \cap C = \{0, 6\}$$

$$A \cap (B \cap C) = \{0, 1, 2, 4, 5\} \cap \{0, 6\}$$

$$A \cap (B \cap C) = \{0\} \dots\dots\dots(3)$$

$A \cap B$ - contains all the common elements in set A and B.

$$A \cap B = \{0, 1, 2, 4, 5\} \cap \{0, 2, 4, 6, 8\}$$

$$A \cap B = \{0, 2, 4\}$$

$$(A \cap B) \cap C = \{0, 2, 4\} \cap \{0, 3, 6, 9\}$$

$$(A \cap B) \cap C = \{0\} \dots\dots\dots(4)$$

Hence from (3) and (4), all the elements of $A \cap (B \cap C)$ and $(A \cap B) \cap C$ are same.

$A \cap (B \cap C) = (A \cap B) \cap C$, i.e., the intersection of sets is associative.

Question 17

If $A = \{x \in W : 5 < x < 10\}$, $B = \{3, 4, 5, 6, 7\}$ and $C = \{x = 2n ; n \in N \text{ and } n \leq 4\}$.

Find :

(i) $A \cap (B \cup C)$

(ii) $(B \cup A) \cap (B \cup C)$

(iii) $B \cup (A \cap C)$

$$(iv) (A \cap B) \cup (A \cap C)$$

Name the sets which are equal.

Answer

$$A = \{x \in W : 5 < x < 10\}$$

$$A = \{6, 7, 8, 9\}$$

$$B = \{3, 4, 5, 6, 7\}$$

$$C = \{x = 2n ; n \in N \text{ and } n \leq 4\}$$

$$C = \{2, 4, 6, 8\}$$

(i) $B \cup C$ - contains all the elements in set B and C.

$$B \cup C = \{3, 4, 5, 6, 7\} \cup \{2, 4, 6, 8\}$$

$$B \cup C = \{2, 3, 4, 5, 6, 7, 8\}$$

$A \cap (B \cup C)$ - contains all the common elements in set A and $(B \cup C)$.

$$A \cap (B \cup C) = \{6, 7, 8, 9\} \cap \{2, 3, 4, 5, 6, 7, 8\}$$

$$\mathbf{A \cap (B \cup C) = \{6, 7, 8\}}$$

(ii) $B \cup A$ - contains all the elements in set B and A.

$$B \cup A = \{3, 4, 5, 6, 7\} \cup \{6, 7, 8, 9\}$$

$$B \cup A = \{3, 4, 5, 6, 7, 8, 9\}$$

$B \cup C$ - contains all the elements in set B and C.

$$B \cup C = \{3, 4, 5, 6, 7\} \cup \{2, 4, 6, 8\}$$

$$B \cup C = \{2, 3, 4, 5, 6, 7, 8\}$$

$(B \cup A) \cap (B \cup C)$ - contains all common elements in set $(B \cup A)$ and $(B \cup C)$.

$$(B \cup A) \cap (B \cup C) = \{3, 4, 5, 6, 7, 8, 9\} \cap \{2, 3, 4, 5, 6, 7, 8\}$$

$$(B \cup A) \cap (B \cup C) = \{3, 4, 5, 6, 7, 8\}$$

(iii) $A \cap C$ - contains all the common elements in set A and C.

$$A \cap C = \{6, 7, 8, 9\} \cap \{2, 4, 6, 8\}$$

$$A \cap C = \{6, 8\}$$

$B \cup (A \cap C)$ - contains all the elements of B and $(A \cap C)$.

$$B \cup (A \cap C) = \{3, 4, 5, 6, 7\} \cup \{6, 8\}$$

$$B \cup (A \cap C) = \{3, 4, 5, 6, 7, 8\}$$

(iv) $A \cap B$ - contains all the common elements in set A and B.

$$A \cap B = \{6, 7, 8, 9\} \cap \{3, 4, 5, 6, 7\}$$

$$A \cap B = \{6, 7\}$$

$A \cap C$ - contains all the common elements in set A and C.

$$A \cap C = \{6, 7, 8, 9\} \cap \{2, 4, 6, 8\}$$

$$A \cap C = \{6, 8\}$$

$(A \cap B) \cup (A \cap C)$ - contains all the elements of $(A \cap B)$ and $(A \cap C)$.

$$(A \cap B) \cup (A \cap C) = \{6, 7\} \cup \{6, 8\}$$

$$(A \cap B) \cup (A \cap C) = \{6, 7, 8\}$$

As we can see that some sets are having same elements i.e, they are equal.

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \text{ and } B \cup (A \cap C) = (B \cup A) \cap (B \cup C)$$

Question 18

If $P = \{\text{factors of } 36\}$ and $Q = \{\text{factors of } 48\}$; find :

(i) $P \cup Q$

(ii) $P \cap Q$

(iii) $Q - P$

(iv) $P' \cap Q$

Answer

$$P = \{\text{factors of } 36\}$$

$$P = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$

$$Q = \{\text{factors of } 48\}$$

$$Q = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$$

(i) $P \cup Q$

$P \cup Q$ - contains all the elements of P and Q .

$$P \cup Q = \{1, 2, 3, 4, 6, 9, 12, 18, 36\} \cup \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$$

$$\mathbf{P \cup Q = \{1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48\}}$$

$$(ii) P \cap Q$$

$P \cap Q$ - contains all the common elements in set P and Q.

$$P \cap Q = \{1, 2, 3, 4, 6, 9, 12, 18, 36\} \cap \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$$

$$\mathbf{P \cap Q = \{1, 2, 3, 4, 6, 12\}}$$

$$(iii) Q - P$$

$Q - P$ - contains all the elements in set Q but not in set P.

$$Q - P = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\} - \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$

$$\mathbf{Q - P = \{8, 16, 24, 48\}}$$

$$(iv) P' \cap Q$$

U is the universal set containing all the elements in set P and Q.

$$U = \{1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48\}$$

P' - contains all the elements of U which are not in P.

$$P' = \{8, 16, 24, 48\}$$

$P' \cap Q$ - contains all the common elements of P' and Q

$$P' \cap Q = \{8, 16, 24, 48\} \cap \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$$

$$\mathbf{P' \cap Q = \{8, 16, 24, 48\}}$$

Question 19

If $A = \{6, 7, 8, 9\}$, $B = \{4, 6, 8, 10\}$ and $C = \{x : x \in \mathbb{N} : 2 < x \leq 7\}$; find :

(i) $A - B$

(ii) $B - C$

(iii) $B - (A - C)$

(iv) $A - (B \cup C)$

(v) $B - (A \cap C)$

(vi) $B - B$

Answer

$$A = \{6, 7, 8, 9\}$$

$$B = \{4, 6, 8, 10\}$$

$$C = \{x : x \in \mathbb{N} : 2 < x \leq 7\}$$

$$C = \{3, 4, 5, 6, 7\}$$

(i) $A - B$

$A - B$ - contains all the elements which are in A but not in B .

$$A - B = \{6, 7, 8, 9\} - \{4, 6, 8, 10\}$$

$$\mathbf{A - B = \{7, 9\}}$$

(ii) $B - C$

$B - C$ - contains all the elements which are in B but not in C .

$$B - C = \{4, 6, 8, 10\} - \{3, 4, 5, 6, 7\}$$

$$\mathbf{B - C = \{8, 10\}}$$

$$(iii) B - (A - C)$$

A - C - contains all the elements which are in A but not in C.

$$A - C = \{6, 7, 8, 9\} - \{3, 4, 5, 6, 7\}$$

$$A - C = \{8, 9\}$$

B - (A - C) - contains all the elements which are in set B but not in (A - C).

$$B - (A - C) = \{4, 6, 8, 10\} - \{8, 9\}$$

$$\mathbf{B - (A - C) = \{4, 6, 10\}}$$

$$(iv) A - (B \cup C)$$

B $\cup$ C - contains all the elements in set B and C.

$$B \cup C = \{4, 6, 8, 10\} \cup \{3, 4, 5, 6, 7\}$$

$$B \cup C = \{3, 4, 5, 6, 7, 8, 10\}$$

A - (B $\cup$ C) - contains all the elements which is in set A but not in (B $\cup$ C).

$$A - (B \cup C) = \{6, 7, 8, 9\} - \{3, 4, 5, 6, 7, 8, 10\}$$

$$\mathbf{A - (B \cup C) = \{9\}}$$

$$(v) B - (A \cap C)$$

A $\cap$ C - contains all the common elements in set A and C.

$$A \cap C = \{6, 7, 8, 9\} \cup \{3, 4, 5, 6, 7\}$$

$$A \cap C = \{6, 7\}$$

$B - (A \cap C)$ - contains all the elements which is in set B but not in $(A \cap C)$.

$$B - (A \cap C) = \{4, 6, 8, 10\} - \{6, 7\}$$

$$\mathbf{B - (A \cap C) = \{4, 8, 10\}}$$

(vi) $B - B$

$B - B$ - contains all the elements that are in set B but not in set B.

$$B - B = \{4, 6, 8, 10\} - \{4, 6, 8, 10\}$$

$$\mathbf{B - B = \phi}$$

Question 20

If $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$; verify :

(i) $A - (B \cup C) = (A - B) \cap (A - C)$

(ii) $A - (B \cap C) = (A - B) \cup (A - C)$.

Answer

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{2, 4, 6, 8\}$$

$$C = \{3, 4, 5, 6\}$$

(i) $A - (B \cup C) = (A - B) \cap (A - C)$

Taking LHS : $A - (B \cup C)$

$B \cup C$ - contains all the elements in set B and C.

$$B \cup C = \{2, 4, 6, 8\} \cup \{3, 4, 5, 6\}$$

$$B \cup C = \{2, 3, 4, 5, 6, 8\}$$

$A - (B \cup C)$ - contains all the elements which are in set A but not in $B \cup C$.

$$A - (B \cup C) = \{1, 2, 3, 4, 5\} - \{2, 3, 4, 5, 6, 8\}$$

$$A - (B \cup C) = \{1\} \dots\dots\dots(1)$$

Taking RHS : $(A - B) \cap (A - C)$

$A - B$ - contains all the elements which are in set A but not in B.

$$A - B = \{1, 2, 3, 4, 5\} - \{2, 4, 6, 8\}$$

$$A - B = \{1, 3, 5\}$$

$A - C$ - contains all the elements which are in set A but not in C.

$$A - C = \{1, 2, 3, 4, 5\} - \{3, 4, 5, 6\}$$

$$A - C = \{1, 2\}$$

$(A - B) \cap (A - C)$ - contains all the common elements in set $(A - B)$ and $(A - C)$.

$$(A - B) \cap (A - C) = \{1, 3, 5\} \cap \{1, 2\}$$

$$(A - B) \cap (A - C) = \{1\} \dots\dots\dots(2)$$

From (1) and (2), we can see

$$\text{LHS} = \text{RHS}$$

$$\text{Hence, } A - (B \cup C) = (A - B) \cap (A - C)$$

$$(ii) A - (B \cap C) = (A - B) \cup (A - C).$$

$$\text{Taking LHS : } A - (B \cap C)$$

$B \cap C$ - contains all the common elements in set B and C.

$$B \cap C = \{2, 4, 6, 8\} \cap \{3, 4, 5, 6\}$$

$$B \cap C = \{4, 6\}$$

$A - (B \cap C)$ - contains all the elements which are in set A but not in $B \cap C$.

$$A - (B \cap C) = \{1, 2, 3, 4, 5\} - \{4, 6\}$$

$$A - (B \cap C) = \{1, 2, 3, 5\} \dots\dots\dots(3)$$

$$\text{Taking RHS : } (A - B) \cup (A - C)$$

$A - B$ - contains all the elements which are in set A but not in B.

$$A - B = \{1, 2, 3, 4, 5\} - \{2, 4, 6, 8\}$$

$$A - B = \{1, 3, 5\}$$

$A - C$ - contains all the elements which are in set A but not in C.

$$A - C = \{1, 2, 3, 4, 5\} - \{3, 4, 5, 6\}$$

$$A - C = \{1, 2\}$$

$(A - B) \cup (A - C)$ - contains all the elements in set $(A - B)$ and $(A - C)$.

$$(A - B) \cup (A - C) = \{1, 3, 5\} \cup \{1, 2\}$$

$$(A - B) \cup (A - C) = \{1, 2, 3, 5\} \dots\dots\dots(4)$$

From (3) and (4), we can see

$$\text{LHS} = \text{RHS}$$

$$\text{Hence, } A - (B \cap C) = (A - B) \cup (A - C)$$

Question 21

Given $A = \{x \in \mathbb{N} : x < 6\}$, $B = \{3, 6, 9\}$ and $C = \{x \in \mathbb{N} : 2x - 5 \leq 8\}$. Show that :

$$(i) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(ii) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Answer

$$A = \{x \in \mathbb{N} : x < 6\}$$

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{3, 6, 9\}$$

$$C = \{x \in \mathbb{N} : 2x - 5 \leq 8\}$$

$$2x - 5 \leq 8$$

$$2x \leq 8 + 5$$

$$2x \leq 13$$

$$x \leq \frac{13}{2}$$

$$x \leq 6.5$$

$$C = \{1, 2, 3, 4, 5, 6\}$$

$$(i) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Taking LHS: $A \cup (B \cap C)$

$B \cap C$ - contains all the common elements in set B and C.

$$B \cap C = \{3, 6, 9\} \cap \{1, 2, 3, 4, 5, 6\}$$

$$B \cap C = \{3, 6\}$$

$A \cup (B \cap C)$ - contains all the elements in set A and $B \cap C$.

$$A \cup (B \cap C) = \{1, 2, 3, 4, 5\} \cup \{3, 6\}$$

$$A \cup (B \cap C) = \{1, 2, 3, 4, 5, 6\} \dots\dots\dots(1)$$

Taking RHS : $(A \cup B) \cap (A \cup C)$

$A \cup B$ - contains all the elements in set A and B.

$$A \cup B = \{1, 2, 3, 4, 5\} \cup \{3, 6, 9\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 9\}$$

$A \cup C$ - contains all the elements in set A and C.

$$A \cup C = \{1, 2, 3, 4, 5\} \cup \{1, 2, 3, 4, 5, 6\}$$

$$A \cup C = \{1, 2, 3, 4, 5, 6\}$$

$(A \cup B) \cap (A \cup C)$ - contains all the common elements in set $(A \cup B)$ and $(A \cup C)$

$$(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5, 6, 9\} \cap \{1, 2, 3, 4, 5, 6\}$$

$$(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5, 6\} \dots\dots\dots(2)$$

From (1) and (2), we get

$$\text{LHS} = \text{RHS}$$

$$\text{Hence, } A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(ii) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Taking LHS: $A \cap (B \cup C)$

$B \cup C$ - contains all the elements in set B and C.

$$B \cup C = \{3, 6, 9\} \cup \{1, 2, 3, 4, 5, 6\}$$

$$B \cup C = \{1, 2, 3, 4, 5, 6, 9\}$$

$A \cap (B \cup C)$ - contains all the common elements in set A and $B \cup C$.

$$A \cap (B \cup C) = \{1, 2, 3, 4, 5\} \cup \{1, 2, 3, 4, 5, 6, 9\}$$

$$A \cap (B \cup C) = \{1, 2, 3, 4, 5\} \dots\dots\dots(3)$$

Taking RHS : $(A \cap B) \cup (A \cap C)$

$A \cap B$ - contains all the common elements in set A and B.

$$A \cap B = \{1, 2, 3, 4, 5\} \cap \{3, 6, 9\}$$

$$A \cap B = \{3\}$$

$A \cap C$ - contains all the common elements in set A and C.

$$A \cap C = \{1, 2, 3, 4, 5\} \cap \{1, 2, 3, 4, 5, 6\}$$

$$A \cap C = \{1, 2, 3, 4, 5\}$$

$(A \cap B) \cup (A \cap C)$ - contains all the elements in set $(A \cap B)$ and $(A \cap C)$

$$(A \cap B) \cup (A \cap C) = \{3\} \cup \{1, 2, 3, 4, 5\}$$

$$(A \cap B) \cup (A \cap C) = \{1, 2, 3, 4, 5\} \dots\dots\dots(4)$$

From (3) and (4), we get

$$\text{LHS} = \text{RHS}$$

$$\text{Hence, } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Question 22

If $n(A) = 30$, $n(B) = 20$ and $n(A \cup B) = 36$, find $n(A \cap B)$.

Answer

$$\text{Given - } n(A) = 30$$

$$n(B) = 20$$

$$n(A \cup B) = 36$$

$$\therefore n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

Putting the values, we get

$$n(A \cap B) = 30 + 20 - 36$$

$$\Rightarrow n(A \cap B) = 50 - 36$$

$$\Rightarrow n(A \cap B) = 14$$

$$\therefore n(A \cap B) = 14$$

Question 23

If $n(A) = 50$, $n(B) = 30$ and $n(A \cap B) = 15$, find $n(A \cup B)$.

Answer

Given - $n(A) = 50$

$$n(B) = 30$$

$$n(A \cap B) = 15$$

$$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Putting the values, we get

$$n(A \cup B) = 50 + 30 - 15$$

$$\Rightarrow n(A \cup B) = 80 - 15$$

$$\Rightarrow n(A \cup B) = 65$$

$$\therefore n(A \cup B) = 65$$

Question 24

If $n(A - B) = 30$, $n(B - A) = 20$ and $n(A \cap B) = 10$, find :

(i) $n(A)$

(ii) $n(B)$

$$(iii) n(A \cup B)$$

Answer

Given:

$$n(A - B) = 30$$

$$n(B - A) = 20$$

$$n(A \cap B) = 10$$

$$(i) n(A - B) = n(A) - n(A \cap B)$$

Putting the values, we get

$$30 = n(A) - 10$$

$$n(A) = 30 + 10$$

$$\mathbf{n(A) = 40}$$

$$(ii) n(B - A) = n(B) - n(A \cap B)$$

Putting the values, we get

$$20 = n(B) - 10$$

$$n(B) = 20 + 10$$

$$\mathbf{n(B) = 30}$$

$$(iii) n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Putting the values, we get

$$n(A \cup B) = 40 + 30 - 10$$

$$\Rightarrow n(A \cup B) = 70 - 10$$

$$\Rightarrow n(A \cup B) = 60$$

Hence, $n(A \cup B) = 60$

Question 25

If $n(A - B) = 30$, $n(B - A) = 48$ and $n(A \cap B) = 15$, find $n(A \cup B)$.

Answer

$$n(A - B) = 30$$

$$n(B - A) = 48$$

$$n(A \cap B) = 15$$

$$\therefore n(A - B) = n(A) - n(A \cap B)$$

Putting the values, we get

$$30 = n(A) - 15$$

$$n(A) = 30 + 15$$

$$n(A) = 45$$

$$\therefore n(B - A) = n(B) - n(A \cap B)$$

Putting the values, we get

$$48 = n(B) - 15$$

$$n(B) = 48 + 15$$

$$n(B) = 63$$

$$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Putting the values, we get

$$n(A \cup B) = 45 + 63 - 15$$

$$\Rightarrow n(A \cup B) = 108 - 15$$

$$\Rightarrow n(A \cup B) = 93$$

$$\therefore n(A \cup B) = 93$$

Test Yourself

Question 1(i)

If $A = \{3, 5, 7\}$ and $B = \{5, 7, 9\}$, then $n(A \cap B)$ is :

1. 6

2. 2

3. 4

4. none of these

Answer

$$A = \{3, 5, 7\}$$

$$B = \{5, 7, 9\}$$

$A \cap B$ - contains all the common elements in set A and B.

$$A \cap B = \{3, 5, 7\} \cap \{5, 7, 9\}$$

$$A \cap B = \{5, 7\}$$

$$n(A \cap B) = 2$$

Hence, option 2 is the correct option.

Question 1(ii)

If $n(\text{universal set}) = 80$ and $n(A) = 50$, then $n(\text{complement of set A})$ is :

1. 80

2. 50

3. 30

4. none of these

Answer

$$n(\text{universal set}) = 80$$

$$n(A) = 50$$

$$n(\text{complement of set } A) = n(\text{universal set}) - n(A)$$

$$\Rightarrow n(\text{complement of set } A) = 80 - 50$$

$$\Rightarrow n(\text{complement of set } A) = 30$$

Hence, option 3 is the correct option.

Question 1(iii)

If $A = \{5, 8, 10\}$ and empty set ϕ . Then $\phi - A$ is equal to :

1. A
2. ϕ
3. $\{0\}$
4. none of these

Answer

$$A = \{5, 8, 10\}$$

$\phi - A$ - contains all the elements in set ϕ but not in A.

$$\phi - A = \phi - \{5, 8, 10\}$$

$$\phi - A = \phi$$

Hence, option 2 is the correct option.

Question 1(iv)

If set $A = \{x : x \in W \text{ and } 0 < x \leq 4\}$, then set A is equal to :

1. $\{0, 1, 2, 3\}$
2. $\{0, 1, 2, 3, 4\}$
3. $\{1, 2, 3, 4\}$
4. none of these

Answer

Set $A = \{x : x \in W \text{ and } 0 < x \leq 4\}$

Set $A = \{1, 2, 3, 4\}$

Hence, option 3 is the correct option.

Question 1(v)

If set $P = \{\text{factors of } 8\}$, then set P is equal to :

1. $\{2, 4, 8\}$
2. $\{1, 2, 4\}$
3. $\{1, 2, 4, 8\}$
4. none of these

Answer

Set $P = \{\text{factors of } 8\}$

Factors of 8 = 1, 2, 4, 8

Set P = {1, 2, 4, 8}

Hence, option 3 is the correct option.

Question 1(vi)

The elements of the set $\{x : x \in \mathbb{Z} \text{ and } x^2 \leq 9\}$ are :

1. 3 and -3
2. $\{-3, -2, -1, 0, 1, 2, 3\}$
3. -2, -1, 0, 1 and 2
4. -3, -2, -1, 0, 1, 2 and 3

Answer

Set = $\{x : x \in \mathbb{Z} \text{ and } x^2 \leq 9\}$

$$x^2 = 0$$

$$x = 0$$

$$x^2 = 1$$

$$x = 1 \text{ or } -1$$

$$x^2 = 4$$

$$x = 2 \text{ or } -2$$

$$x^2 = 9$$

$$x = 3 \text{ or } -3$$

$$\text{Set} = \{-3, -2, -1, 0, 1, 2, 3\}$$

Hence, option 2 is the correct option.

Question 1(vii)

If A and B are two equal sets, then $A - B$ is equal to :

1. A

2. B

3. $\{0\}$

4. $\{ \}$

Answer

$A - B$ - contains all the elements in set A but not in B.

A and B are equal set means all the elements in set A and B are same.

$$A - B = \{ \}$$

Hence, option 4 is the correct option.

Question 1(viii)

If $n(A) = n(B)$ then :

1. $A = B$

2. $A \neq B$

3. $A - B = \{0\}$

4. none of these

Answer

$$n(A) = n(B)$$

That means, number of elements in set A and B are same but elements can be different. Hence, none of the option is always correct.

Hence, option 4 is the correct option.

Question 1(ix)

A set has 5 elements, then number of its subsets is :

1. 2^5

2. 5^2

3. $2^5 - 1$

4. 2×5

Answer

If a set has n elements, the number of subsets = 2^n

If a set has 5 elements, the number of subsets = 2^5

Hence, option 1 is the correct option.

Question 1(x)

Let $M = \{\text{factors of } 12\}$ and $N = \{\text{factors of } 24\}$ then $\{24\}$ is equal to :

1. $M \cup N$

2. $M \cap N$

3. $M - N$

4. $N - M$

Answer

$$M = \{\text{factors of } 12\}$$

$$\text{Factors of } 12 = 1, 2, 3, 4, 6, 12$$

$$M = \{1, 2, 3, 4, 6, 12\}$$

$$N = \{\text{factors of } 24\}$$

$$\text{Factors of } 24 = 1, 2, 3, 4, 6, 8, 12, 24$$

$$N = \{1, 2, 3, 4, 6, 8, 12, 24\}$$

$$M \cup N = \{1, 2, 3, 4, 6, 12\} \cup \{1, 2, 3, 4, 6, 8, 12, 24\}$$

$$= \{1, 2, 3, 4, 6, 8, 12, 24\}$$

$$M \cap N = \{1, 2, 3, 4, 6, 12, 24\} \cap \{1, 2, 3, 4, 6, 8, 12, 24\}$$

$$= \{1, 2, 3, 4, 6, 12\}$$

$$M - N = \{1, 2, 3, 4, 6, 12\} - \{1, 2, 3, 4, 6, 8, 12, 24\}$$

$$= \{\}$$

$$N - M = \{1, 2, 3, 4, 6, 8, 12, 24\} - \{1, 2, 3, 4, 6, 12\}$$

$$= \{8, 24\}$$

Hence, none of the given options are correct.

Question 1(xi)

Statement 1: The number of subsets of $\{\{1, \{0\}\}, 2\}$ is 8.

Statement 2: A set containing 'n' elements has $2^n - 1$ proper subsets.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

Given set : $\{\{1, \{0\}\}, 2\}$

The set $A = \{\{1, \{0\}\}, 2\}$ has two elements : $\{1, \{0\}\}$ and 2.

Thus, $n = 2$.

The number of subsets of a set with n elements $= 2^n = 2^2 = 4$.

Thus, statement 1 is false.

We know that,

The total number of proper subsets of a set with n elements is $2^n - 1$.

Thus, statement 2 is true.

Hence, option 4 is the correct option.

Question 1(xii)

Assertion (A) : Let $A = \{1, \{\Phi\}\}$, then each of Φ , $\{1\}$, $\{\{\Phi\}\}$ is a proper subset of A .

Reason (R) : The empty set has no proper subset.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given set $A = \{1, \{\Phi\}\}$

Proper subset of $A = \Phi, \{1\}, \{\{\Phi\}\}$

So, assertion (A) is true.

The only subset of empty set is itself, so we can say that empty set does not have a proper subset.

So, reason (R) is true but reason does not explain assertion.

Hence, option 2 is the correct option.

Question 1(xiii)

Assertion (A) : Let $A = \{\text{factors of } 12\}$ and $B = \{\text{factors of } 16\}$. Then $B - A = \{8, 16\}$

Reason (R) : $B - A = \{x \mid x \in A, \text{ but } x \notin B\}$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

$$\Rightarrow A = \{\text{factors of } 12\}$$

$$\Rightarrow A = \{1, 2, 3, 4, 6, 12\}$$

$$\Rightarrow B = \{\text{factors of } 16\}$$

$$\Rightarrow B = \{1, 2, 4, 8, 16\}$$

Set $B - A$ contains all the elements that are in set B but not in set A.

$$\Rightarrow B - A = \{x \mid x \in B, \text{ but } x \notin A\}$$

So, reason (R) is false.

$$\Rightarrow B - A = \{8, 16\}$$

So, assertion (A) is true.

Hence, option 3 is the correct option.

Question 1(xiv)

Assertion (A) : Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 3, 5, 7, 9\}$ then $A \cap B \subseteq A$ and $A \cap B \subseteq B$, always true for every pair of two sets.

Reason (R) : For any sets A and B, we have $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given : $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 3, 5, 7, 9\}$

$A \cap B$ means this set contains elements common to both A and B.

$$A \cap B = \{1, 3, 5\}$$

$A \cap B \subseteq A$ means $\{1, 3, 5\}$ is subset of $\{1, 2, 3, 4, 5, 6\}$, which is true.

$A \cap B \subseteq B$ means $\{1, 3, 5\}$ is subset of $\{1, 3, 5, 7, 9\}$, which is also true.

So, assertion (A) is true.

For any sets A and B, we have $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

This is a fundamental property of set theory :

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

So any element of $A \cap B$ is automatically in both A and B, which implies $A \cap B \subseteq A$ and $A \cap B \subseteq B$ are true.

So, reason (R) is true and it clearly explains assertion.

Hence, option 1 is the correct option.

Question 1(xv)

Assertion (A) : Let $A = \{x \mid x + 3 = 0, x \in \mathbb{N}\}$, $B = \{x \mid x \leq 3, x \in \mathbb{W}\}$ then $A \cap B = B$.

Reason (R) : For any set A, $A \cap \Phi = \Phi$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given : $A = \{x \mid x + 3 = 0, x \in \mathbb{N}\}$, $B = \{x \mid x \leq 3, x \in \mathbb{W}\}$

$$\Rightarrow x + 3 = 0$$

$$\Rightarrow x = -3$$

Since, $x \in \mathbb{N}$. So, $A = \{\}$

Whole number = $\{0, 1, 2, 3, \dots\}$ and $x \leq 3$

So, $B = \{0, 1, 2, 3\}$

$A \cap B$ means this set contains elements common to both A and B.

$$\Rightarrow A \cap B = \{\}$$

$$\Rightarrow A \cap B \neq B$$

So, assertion (A) is false.

For any set A, $A \cap \Phi = \Phi$

The intersection of any set with the empty set is always the empty set.

So, reason (R) is true.

Hence, option 4 is the correct option.

Question 2

If universal set = {all digits in our number system} and set A = {2, 3, 7, 9}. Write the complement of set A.

Answer

Universal set = {all digits in our number system}

Universal set = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

Set A = {2, 3, 7, 9}

Complement of set A = Universal set - Set A

Complement of set A = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9} - {2, 3, 7, 9}

Complement of set A = {0, 1, 4, 5, 6, 8}

Question 3

If $A = \{\text{factors of } 36\}$ and $B = \{\text{factors of } 48\}$, find :

(i) $A \cup B$

(ii) $A \cap B$

(iii) $A - B$

(iv) $B - A$

Answer

$A = \{\text{factors of } 36\}$

Factors of 36 = 1, 2, 3, 4, 6, 9, 12, 18, 36

$A = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$

$B = \{\text{factors of } 48\}$

Factors of 48 = 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

$B = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$

(i) $A \cup B$

$A \cup B$ - contains all the elements in set A and B.

$A \cup B = \{1, 2, 3, 4, 6, 9, 12, 18, 36\} \cup \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$

$A \cup B = \{1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48\}$

(ii) $A \cap B$

$A \cap B$ - contains all the common elements in set A and B.

$$A \cap B = \{1, 2, 3, 4, 6, 9, 12, 18, 36\} \cap \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$$

$$\mathbf{A \cap B = \{1, 2, 3, 4, 6, 12\}}$$

(iii) $A - B$

$A - B$ - contains all the elements in set A but not in B.

$$A - B = \{1, 2, 3, 4, 6, 9, 12, 18, 36\} - \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$$

$$\mathbf{A - B = \{9, 18, 36\}}$$

(iv) $B - A$

$B - A$ - contains all the elements in set B but not in A.

$$B - A = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\} - \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$

$$\mathbf{B - A = \{8, 16, 24, 48\}}$$

Question 4

By taking the sets of your own, verify that :

$$(i) n(A - B) = n(A \cup B) - n(B)$$

$$(ii) n(A \cap B) + n(A \cup B) = n(A) + n(B)$$

Answer

Lets take the set $A = \{1, 2, 3, 4, 5, 6\}$

and set $B = \{2, 4, 6\}$

$$(i) n(A - B) = n(A \cup B) - n(B)$$

$A - B$ - contains all the elements in set A but not in B.

$$A - B = \{1, 2, 3, 4, 5, 6\} - \{2, 4, 6\}$$

$$A - B = \{1, 3, 5\}$$

$$n(A - B) = 3$$

$A \cup B$ - contains all the elements in set A and B.

$$A \cup B = \{1, 2, 3, 4, 5, 6\} \cup \{2, 4, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$n(A \cup B) = 6$$

$$n(B) = 3$$

Taking LHS : $n(A - B)$

$$n(A - B) = 3$$

Taking RHS : $n(A \cup B) - n(B)$

$$n(A \cup B) - n(B) = 6 - 3$$

$$n(A \cup B) - n(B) = 3$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore \mathbf{n(A - B) = n(A \cup B) - n(B)}$$

$$(ii) \ n(A \cap B) + n(A \cup B) = n(A) + n(B)$$

$A \cap B$ - contains all the common elements in set A and B.

$$A \cap B = \{1, 2, 3, 4, 5, 6\} \cap \{2, 4, 6\}$$

$$A \cap B = \{2, 4, 6\}$$

$$n(A \cap B) = 3$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$n(A \cup B) = 6$$

$$n(A) = 6$$

$$n(B) = 3$$

Taking LHS:

$$n(A \cap B) + n(A \cup B)$$

$$n(A \cap B) + n(A \cup B) = 3 + 6$$

$$n(A \cap B) + n(A \cup B) = 9$$

Taking RHS:

$$n(A) + n(B)$$

$$n(A) + n(B) = 6 + 3$$

$$n(A) + n(B) = 9$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore n(A \cap B) + n(A \cup B) = n(A) + n(B)$$

Question 5

If $n(A - B) = 24$, $n(B - A) = 32$ and $n(A \cap B) = 10$; find $n(A \cup B)$.

Answer

$$n(A - B) = 24$$

$$n(B - A) = 32$$

$$n(A \cap B) = 10$$

$$\therefore n(A - B) = n(A) - n(A \cap B)$$

Putting the values, we get

$$24 = n(A) - 10$$

$$n(A) = 24 + 10$$

$$n(A) = 34$$

$$\therefore n(B - A) = n(B) - n(A \cap B)$$

Putting the values, we get

$$32 = n(B) - 10$$

$$n(B) = 32 + 10$$

$$n(B) = 42$$

$$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Putting the values, we get

$$n(A \cup B) = 34 + 42 - 10$$

$$\Rightarrow n(A \cup B) = 76 - 10$$

$$\Rightarrow n(A \cup B) = 66$$

$$\therefore n(A \cup B) = 66$$

Question 6

If $\xi = \{x : x \in \mathbb{N}, x \leq 10\}$, $A = \{x : x \geq 5\}$ and $B = \{x : 3 < x < 6\}$, then find :

(i) $(A \cup B)'$

(ii) $A' \cap B'$

Are $(A \cup B)'$ and $A' \cap B'$ equal ?

Answer

$$\xi = \{x : x \in \mathbb{N}, x \leq 10\}$$

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{x : x \geq 5\}$$

$$A = \{5, 6, 7, 8, 9, 10\}$$

$$B = \{x : 3 < x < 6\}$$

$$B = \{4, 5\}$$

(i) $(A \cup B)'$

$A \cup B$ - contains all the elements in set A and B.

$$A \cup B = \{5, 6, 7, 8, 9, 10\} \cup \{4, 5\}$$

$$A \cup B = \{4, 5, 6, 7, 8, 9, 10\}$$

$(A \cup B)'$ - contains all the elements in universal set which are not in $A \cup B$.

$$(A \cup B)' = \xi - A \cup B$$

$$(A \cup B)' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{4, 5, 6, 7, 8, 9, 10\}$$

$$(A \cup B)' = \{1, 2, 3\} \dots\dots\dots(1)$$

$$(A \cup B)' = \{1, 2, 3\}$$

$$(ii) A' \cap B'$$

A' - contains all the elements in universal set which are not in A .

$$A' = \xi - A$$

$$A' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{5, 6, 7, 8, 9, 10\}$$

$$A' = \{1, 2, 3, 4\}$$

B' - contains all the elements in universal set which are not in B .

$$B' = \xi - B$$

$$B' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{4, 5\}$$

$$B' = \{1, 2, 3, 6, 7, 8, 9, 10\}$$

$A' \cap B'$ - contains all the common elements in set A and B .

$$A' \cap B' = \{1, 2, 3, 4\} \cap \{1, 2, 3, 6, 7, 8, 9, 10\}$$

$$A' \cap B' = \{1, 2, 3\} \dots\dots\dots(2)$$

$$A' \cap B' = \{1, 2, 3\}$$

From (1) and (2), we can see that

Yes, $(A \cup B)'$ and $A' \cap B'$ are equal.

Question 7

Write the elements of the set

$$\{x : x = 3y - 1, y \in \mathbb{N} \text{ and } 8 < y \leq 12\}.$$

Answer

$$\{x : x = 3y - 1, y \in \mathbb{N} \text{ and } 8 < y \leq 12\}$$

$$y = 9$$

$$x = 3y - 1 = 3 \times 9 - 1$$

$$x = 27 - 1 = 26$$

$$y = 10$$

$$x = 3y - 1 = 3 \times 10 - 1$$

$$x = 30 - 1 = 29$$

$$y = 11$$

$$x = 3y - 1 = 3 \times 11 - 1$$

$$x = 33 - 1 = 32$$

$$y = 12$$

$$x = 3y - 1 = 3 \times 12 - 1$$

$$x = 36 - 1 = 35$$

Hence, the elements are 26, 29, 32 and 35

Question 8

If universal set = $\{x : x \in \mathbb{Z}, -2 \leq x < 4\}$,

$A = \{x : -1 \leq x < 3\}$, $B = \{x : 0 < x < 4\}$ and

$C = \{x : -2 \leq x \leq 0\}$; show that :

$$A - (B \cup C) = (A - B) \cap (A - C)$$

Answer

Universal set = $\{x : x \in \mathbb{Z}, -2 \leq x < 4\}$

Universal set = $\{-2, -1, 0, 1, 2, 3\}$

$A = \{x : -1 \leq x < 3\}$

$A = \{-1, 0, 1, 2\}$

$B = \{x : 0 < x < 4\}$

$B = \{1, 2, 3\}$

$C = \{x : -2 \leq x \leq 0\}$

$C = \{-2, -1, 0\}$

To prove:

$$A - (B \cup C) = (A - B) \cap (A - C)$$

Taking LHS:

$$A - (B \cup C)$$

$B \cup C$ - contains all the elements in set B and C.

$$B \cup C = \{1, 2, 3\} \cup \{-2, -1, 0\}$$

$$B \cup C = \{-2, -1, 0, 1, 2, 3\}$$

$A - (B \cup C)$ - contains all the elements in set A but not in $B \cup C$.

$$A - (B \cup C) = \{-1, 0, 1, 2\} - \{-2, -1, 0, 1, 2, 3\}$$

$$A - (B \cup C) = \{ \}$$

Taking RHS:

$$(A - B) \cap (A - C)$$

$A - B$ - contains all the elements in set A but not in B.

$$A - B = \{-1, 0, 1, 2\} - \{1, 2, 3\}$$

$$A - B = \{-1, 0\}$$

$A - C$ - contains all the elements in set A but not in C.

$$A - C = \{-1, 0, 1, 2\} - \{-2, -1, 0\}$$

$$A - C = \{1, 2\}$$

$(A - B) \cap (A - C)$ - contains all the common elements in set $(A - B)$ and $(A - C)$.

$$(A - B) \cap (A - C) = \{-1, 0\} \cap \{1, 2\}$$

$$(A - B) \cap (A - C) = \{\}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore \mathbf{A - (B \cup C) = (A - B) \cap (A - C)}$$

Question 9

Let set $A = \{x : x \in \mathbb{Z} \text{ and } x^2 - 9 = 0\}$ and set $B = \{x : x \in \mathbb{W} \text{ and } x^2 - 16 < 0\}$; then find :

(i) $A \cup B$

(ii) $B \cap A$

Answer

$$\text{Set } A = \{x : x \in \mathbb{Z} \text{ and } x^2 - 9 = 0\}$$

$$x^2 - 9 = 0$$

$$x^2 = 9$$

$$x = \sqrt{9}$$

$$x = 3 \text{ or } -3$$

$$\text{Set } A = \{-3, 3\}$$

$$\text{Set } B = \{x : x \in \mathbb{W} \text{ and } x^2 - 16 < 0\}$$

$$x^2 - 16 < 0$$

$$x^2 < 16$$

$$x < \sqrt{16}$$

$$x < 4$$

$$\text{Set B} = \{0, 1, 2, 3\}$$

$$(i) A \cup B$$

$A \cup B$ - contains all the elements of set A and B.

$$A \cup B = \{-3, 3\} \cup \{0, 1, 2, 3\}$$

$$\mathbf{A \cup B = \{-3, 0, 1, 2, 3\}}$$

$$(ii) B \cap A$$

$B \cap A$ - contains all the common elements in set B and A.

$$B \cap A = \{0, 1, 2, 3\} \cap \{-3, 3\}$$

$$\mathbf{B \cap A = \{3\}}$$

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